

SMART BLOOD DONATION MANAGEMENT SYSTEM

Abstract

Blood donation plays a vital role in healthcare, yet many existing blood bank systems rely on manual processes that lead to delays in donor identification, inefficient record management, and difficulty in responding to urgent blood requests. This project, **"AI-Enabled Smart Blood Donation Management System Using the MERN Stack,"** presents a web-based solution designed to streamline the management of blood donors, recipients, blood requests, and donation records. The system is developed using the MERN stack, comprising MongoDB for secure data storage, Express.js and Node.js for backend services, and React.js for a responsive and interactive user interface. The application enables user registration, donor profile management, blood request submission, request tracking, and administrative monitoring through a centralized platform.

The system incorporates AI-enabled features within the MERN application to enhance donor management and improve the efficiency of blood request handling. Based on stored donor information such as blood group, availability status, and location, the system assists in identifying suitable donors for emergency requests, thereby reducing response time and improving coordination between donors and recipients. By digitizing the complete blood donation workflow, the proposed system minimizes manual effort, improves data accuracy, and provides a secure, user-friendly, and scalable platform for effective blood donation management. The project demonstrates how AI-assisted decision support integrated with the MERN stack can contribute to a more efficient and reliable blood donation ecosystem.